MIL-M-18598(NOrd)

11 May 1955
SUPERSEDING
NAVORD OS 2825 C
22 October 1953

MILITARY SPECIFICATION

MARKER, DEPTH CHARGE,MK 2 MOD 0

(FOR NIGHT USE)

1. SCOPE

1.1 This specification governs the manufacture, assembly, and preparation for delivery of Depth Charge Marker Mk 2 Mod 0 (For Night Use), and the methods of inspection and tests upon which acceptance will be based. The marker is launched by removing the tear strips at both ends by use of the rings, and throwing the marker overboard. Water enters the holes at both ends and reacts with the calcium carbide to produce flammable acetylene gas and with the calcium phosphide to produce spontaneously flammable phosphide gas. Both gases are emitted from the small holes in the top of the container, which floats in an upright position, to produce a luminous flame within a period of two minutes after striking the water.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN-P-108 -- Packaging and Packing for Overseas Shipment Boxes, Fiberboard (V-board and W-board), Exterior and Interior.

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STANDARDS

MILITARY

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 -- Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDNANCE

LD SK 102927 -- Depth Charge Marker Mk 2 Mod 0 (For Night Use),
LD 261610 -- Depth Charge Marker Container Mk 1 Mod 0,
And all documents listed on the above LD's.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the Procuring Activity or as directed by the Contracting Officer.)

2.2 Other publications. -- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 -- Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950.")

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3. REQUIREMENTS

3.1 General.

3.1.1 The depth charge marker covered by this specification shall be manufactured in accordance with Bureau of Ordnance List of Drawings LD SK No. 102927 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Chemicals. -- The container shall be filled with sufficient calcium carbide and calcium phosphide to operate the marker for at least 30 minutes.

3.3 Soldering. -- All soldering shall produce watertight joints and shall be in accordance with standards of good commercial practice.

3.4 Functioning. -- Each lot of markers shall pass the functioning test specified in 4.5.

3.5 Workmanship. -- Only the best trade practices shall be used in the manufacture of the Depth Charge Marker Mk 2 Mod 0 (For Night Use). Care shall be exercised in the assembly of the component parts to assure the proper seating and alignmen especially where crimping is employed to maintain parts in their relative positions. There shall be no burrs, cracks, or other imperfections of manufacture which would harmfully affect the assembly or functioning of the product. All parts shall be free of chips, dirt, grease and other foreign material. The cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent which is employed.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of Standard MIL-STD-105.

4.1.1 Lot. -- As applied to the Government Inspection of units of product, the term "Lot" shall mean "Inspection Lot", i.e., a collection of units of the product used as a basis for Government Inspection. Unless otherwise specified, the number of units of product in "Inspection Lots" shall be as determined

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by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Preproduction sample. -- Before entering into quantity production, a preproduction sample of twelve (12) Depth Charge Markers Mk 2 Mod 0 (For Night Use) shall be delivered by the Contractor for test to an agency designated by the Procuring Agency. These markers shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for a subsequent contract. -- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Production lot. -- A production lot shall consist of at least 1,000 markers or one weeks production (not to exceed 4,000) whichever is greater. Unless otherwise stated in the contract or order, all units for test purposes shall be at the expense of the Contractor. The production lot should not be confused with the "inspection lot" as defined in Standard MIL-STD-105.

4.4 Inspection. -- The general inspection procedures shall be in accordance with "General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment", OP 400; Government Inspection shall be in accordance with the provisions of General Procedures for Government Inspection and Contractor's Inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.5 Functioning test. -- Select and test samples from each lot in accordance with Multiple Sampling Plan, Inspection Level I, AQL 10.0 percent as prescribed by Standard MIL-STD-105.

4.5.1 Test.

4.5.1.1 The markers shall be launched at an altitude of 1,000 feet.

4.5.1.2 The markers shall return to the surface of the water and ignite within two minutes, and then burn for at least 30 minutes.

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4.5.1.3 If a marker should become submerged by wave action during its burning, it shall return to the surface and reignite.

4.5.2 Acceptability. -- Failures shall be defined as specified in 4.5.2.1 through 4.5.2.3.

4.5.2.1 Failure of the marker to ignite within two minutes after initial contact with water.

4.5.2.2 Failure of the marker to burn for at least 30 minutes.

4.5.2.3 Failure of the marker to surface and reignite if it becomes submerged by wave action.

4.5.3 Test report. -- Test reports shall be submitted to an agency designated by the Government.

4.5.3.1 All test reports shall include the observed burning time, the approximate height of flame produced, if possible, and the delay time before ignition of marker.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.

5.1.1 Fifteen markers shall be packaged and packed in a corrugated fiberboard carton conforming to the requirements of Specification JAN-P-108, W5C with separators conforming to the same specification.

5.1.2 When specifically ordered, packaging shall be in accordance with the requirements of Bureau of Ordnance List of Drawings LD 261610, Depth Charge Marker Container Mk 1 Mod 0.

5.2 Marking. -- In addition to any special marking required by the contract or order, all containers shall be marked in compliance with Standard MIL-STD-129 and the Code of Federal Regulations 49 CFR 71-78.

5.3 Manufacturing data cards.

5.3.1 For Department of the Navy purchases, three copies of Pyrotechnic Ammunition Manufacturing Data Cards (NavOrd Form 1204), furnished by the Government shall be filled out by the manufacturer except for the Acceptance Data Section.

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5.3.1.1 The following items shall be listed under the Component Section:

Calcium Carbide
Calcium Phosphide

5.3.1.2 All deviations from specifications and drawings shall be listed on the bottom of the card.

5.3.1.3 The Government Inspector shall fill in the Acceptance Data Section, endorse the cards, and forward them to the Bureau of Ordnance.

5.4 Performance data cards.

5.4.1 For Department of the Navy purchases, two Pyrotechnic Ammunition Performance Data Cards (NAVORD Form 1187), furnished by the Government, shall be filled out by the manufacturer and packed with each ten markers, and secured in such a manner that the forms will go with the signals to the point of use.

5.4.1.1 The form shall have the following information;
Name of Pyrotechnic Ammunition - Depth Charge Marker, Night, Mk 2 Mod 0.
Contract No. _____(fill in number of contract)
Date of Manufacture _____(fill in date of final assembly)
Assembly Lot No. _____(fill in particular lot number)
Under Reference (b) _____(fill in OP No. 1177 and OP No. 998)

5.4.1.2 Below the line titled "Date Fired" enter the following:

Tear Strip Action
Ignition
Burning Time (45 min.)
Flame Size

5.4.1.3 The Schedule of Performance Surveillance Section on the rear of the form shall be filled in as specified in 5.4.1.3.1 and 5.4.1.3.2.

5.4.1.3.1 Before the word "rounds" insert the figure "1".

5.4.1.3.2 After the word "minimum" insert the phrase "about 1 April and 1 October".

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6. NOTES

6.1 Ordering data. -- Procurement documents should specify the following:

(a) Title, symbol and date of this specification.
(b) Type of packing required (see 5.1.1 and 5.1.2).

Patent notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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